WORKED SOLUTION

STEP III 2018, Question 6

Pure • Complex Numbers and Geometry

STEP III 2018, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q6

The key idea

A complex number is real exactly when it equals its own conjugate, so collinearity of three points turns into an equation between z's and z^*'s. On the unit circle, $a^* = 1/a$, which lets us clear the conjugates of the circle points and leaves a *linear* equation for any point q on a chord. Each later result comes from writing down two such chord equations and adding or subtracting them.

Solution

Part (i)

Since A, Q, C are collinear and $A \neq C$, the point Q is A plus a real multiple of the vector from A to C: $q - a = t(c - a)$ for some real t. As $c \neq a$ we may divide, so

$$\frac{q-a}{c-a} = t \quad \text{is real.}$$

A real number equals its conjugate, so

$$\frac{q-a}{c-a} = \frac{q^*-a^*}{c^*-a^*},$$

where $c^* - a^* \neq 0$ because $c \neq a$. Cross-multiplying gives $(c-a)(q^*-a^*) = (c^*-a^*)(q-a)$, as required.

Now suppose $aa^* = cc^* = 1$, so $a^* = 1/a$ and $c^* = 1/c$. Then

$$c^* - a^* = \frac{1}{c} - \frac{1}{a} = \frac{a-c}{ac} = -\frac{c-a}{ac},$$

and the previous equation becomes

$$(c-a)\left(q^* - \frac{1}{a}\right) = -\frac{c-a}{ac}(q-a).$$

Dividing by $c - a \neq 0$ and multiplying by $-ac$,

$$-ac\,q^* + c = q - a, \qquad \text{that is,} \qquad q + ac\,q^* = a + c.$$

This is the equation of the chord AC: every point q on the line through A and C satisfies it.

Part (ii)

Because A, B, C, D are in anticlockwise order, AC and BD are the diagonals of a convex quadrilateral inscribed in the circle, so they cross at a point Q strictly inside the circle. In particular Q is different from all four points, so part (i) applies to the lines AC and BD:

$$q + ac\,q^* = a + c,$$
$$q + bd\,q^* = b + d.$$

Subtracting the second from the first removes q:

$$(ac - bd)\,q^* = (a + c) - (b + d). \tag{1}$$

? **Ask yourself** We have q^*; how do we get an equation for q to add to it?

Take the conjugate of (1) and again use $a^* = 1/a$ and so on. The left side becomes

$$\Big(\frac{1}{ac} - \frac{1}{bd}\Big)q = \frac{bd - ac}{abcd}\,q,$$

and the right side becomes

$$\frac{1}{a} + \frac{1}{c} - \frac{1}{b} - \frac{1}{d} = \frac{a + c}{ac} - \frac{b + d}{bd} = \frac{bd(a + c) - ac(b + d)}{abcd}.$$

Multiplying by $-abcd$,

$$(ac - bd)\,q = ac(b + d) - bd(a + c). \tag{2}$$

Adding (1) and (2),

$$(ac - bd)(q + q^*) = a + c - b - d + abc + acd - abd - bcd.$$

Grouping the terms on the right,

$$(a - b) + (acd - bcd) + (c - d) + (abc - abd) = (a - b)(1 + cd) + (c - d)(1 + ab),$$

which gives the required result.

Part (iii)

The lines AB and CD are opposite sides of the inscribed quadrilateral, so where they meet, at P, is outside the circle. So P is distinct from A and B, and from C and D. Applying part (i) to the collinear points A, P, B gives $p + ab\,p^* = a + b$. Since p is real, $p^* = p$, and so

$$p(1 + ab) = a + b.$$

In the same way, from C, P, D,

$$p(1 + cd) = c + d.$$

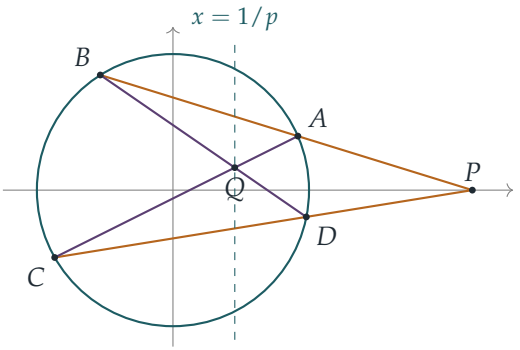
Now multiply the final result of part (ii) by p and substitute these two equations:

$$\begin{aligned} p\,(ac - bd)(q + q^*) &= (a - b)\,p(1 + cd) + (c - d)\,p(1 + ab) \\ &= (a - b)(c + d) + (c - d)(a + b) \\ &= (ac + ad - bc - bd) + (ac + bc - ad - bd) \\ &= 2(ac - bd). \end{aligned}$$

Since $ac - bd \neq 0$, we may divide by it, giving $p(q + q^*) = 2$, as required.

In fact $ac - bd$ can never be zero here. If $ac = bd$, then (1) forces $a + c = b + d$ as well, so $z^2 - (a + c)z + ac$ and $z^2 - (b + d)z + bd$ are the same quadratic. Then $\{a, c\} = \{b, d\}$, which contradicts the points being distinct.

Since $q + q^* = 2\,\mathrm{Re}\,q$, the result says $\mathrm{Re}\,q = 1/p$. However the two lines through P are chosen, the crossing point Q of the diagonals lies on the fixed vertical line $x = 1/p$.



Checks

- Symmetric case: take A, B at angles $20°$, $70°$ and C, D at $-70°$, $-20°$. By the reflection in the real axis, P and Q are both real, and the result says $pq = 1$. With $a = \mathrm{e}^{\mathrm{i}\alpha}$, $b = \mathrm{e}^{\mathrm{i}\beta}$, the formula $p = (a + b)/(1 + ab)$ gives $p = \cos\frac{\alpha-\beta}{2} / \cos\frac{\alpha+\beta}{2} = \cos 25° / \cos 45° \approx 1.2817$. Find q directly as the point where AC crosses the real axis: $q \approx 0.7802$, and $pq \approx 1.0000$.

- The diagram uses $p = 2.2$. The diagonals cross at $x \approx 0.4545 = 1/2.2$, as predicted. All three identities were also confirmed by computer on thousands of random configurations.

Common slips

- Forgetting that conjugating $ac - bd$ gives $1/(ac) - 1/(bd)$, not $ac - bd$ again; the factor $abcd$ must be cleared carefully.
- Dividing by $c - a$ or $ac - bd$ without saying why it is non-zero.

Beyond STEP

Here Q lies on the *polar* of P with respect to the circle: for a point P outside the unit circle at distance p from the centre, its polar is the line perpendicular to OP at distance $1/p$ from O. This is part of Brocard's theorem in projective geometry, and the map $P \mapsto 1/p^*$ behind it is *inversion* in the circle, a key tool in complex analysis and geometry. Needham's *Visual Complex Analysis* develops inversion and the geometry of the Argand diagram beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.